

Benchmarking of NQCH's quantum computer

April 25, 2026

1. Report of Changes

Platform: sinq20
Calibration-id: 287d9f2de17a0c2e70ff663d95a723c1b895e392
Calibration date: 2026-04-23 10:25:04
Calibration note: chore(sinq20): readout retuning after twpa pump change

Experiment-id: 20260425212203
Experiment date: 2026-04-25 21:22:04
Experiment note: temporary note!!!

Platform: sinq20
Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78
Calibration date: 2026-01-15 02:03:03
Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538
Experiment date: 2026-01-18 16:15:38
Experiment note: temporary note!!!

2. Version Comparison

Library	Version	Library	Version
qibo	0.2.23	numpy	2.4.4
qibolab	0.2.9	qibocal	0.2.5
matplotlib	3.10.8	scipy	1.17.1
scikit-learn	1.8.0	pandas	2.3.3
networkx	3.6.1	sympy	1.14.0
torch	2.11.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



4. Statistics

	Average	Median	Min	Max
T1 (ns)	21.3	21.3	13.5	30
T2 (ns)	11.2	10.2	1.86	26.8
Fidelity	0.994	0.998	0.98	0.999
RO Fidelity	0.958	0.959	0.94	0.969
2q RB Fidelity	-	-	-	-
2q CZ Fidelity	-	-	-	-

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	0.99	0.997	0.887	0.999
RO Fidelity	0.794	0.777	0.777	0.927
2q RB Fidelity	0.965	0.972	0.923	0.981
2q CZ Fidelity	0.975	0.974	0.967	0.991

5. Best Qubits Selection

k-qubits	Best Qubits	Fidelity
2	8, 9	0.915
3	3, 8, 9	0.914
4	3, 8, 9, 13	0.913
5	3, 8, 9, 12, 13	0.912

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

6. Summary metrics

Benchmark	Metric	Value	Runtime
Mermin	Max	1.08	235.91 sec
Grover 2q	Max prob	0.839	4.89 sec
Grover 3q	Max prob	0.265	2.95 sec
GHZ state prep.	Success	0.497	3.33 sec

Benchmark	Metric	Value	Runtime
Mermin	Max	-3.316	1.02 sec
Grover 2q	Max prob	0.89	2.36 sec
Grover 3q	Max prob	0.385	2.43 sec
GHZ state prep.	Success	0.910	3.95 sec

7. Randomized Benchmarking Results

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.995	± 0.000224	10	0.996	± 0.00012
1	0.991	± 0.000662	11	0.997	± 0.000156
2	0.998	± 8.9e-05	12	0.998	± 7.26e-05
3	0.998	± 7.97e-05	13	0.998	± 9.89e-05
4	0.991	± 0.000559	14	0.998	± 5.64e-05
5	0.997	± 0.000197	15	0.998	± 0.000118
6	0.994	± 0.00018	16	0.982	± 0.00376
7	0.976	± 0.00421	17	0.956	± 0.0121
8	0.998	± 6.71e-05	18	0.995	± 0.000286
9	0.998	± 8.32e-05	19	0.996	± 0.000122

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131	10	0.989	± 0.000582
1	0.998	± 7.47e-05	11	0.996	± 0.000218
2	0.998	± 7.57e-05	12	0.998	± 0.000126
3	0.995	± 0.000163	13	0.998	± 7.15e-05
4	0.996	± 0.000185	14	0.985	± 0.00183
5	0.995	± 0.000147	15	0.995	± 0.000469
6	0.998	± 8.34e-05	16	0.993	± 0.000558
7	0.887	± 0.0287	17	0.99	± 0.000502
8	0.999	± 4.48e-05	18	0.997	± 0.00026
9	0.998	± 7.2e-05	19	0.998	± 9.05e-05

8. Mermin

Mermin's algorithm for 3 qubits.

Runtime	Qubits	Max
235.91 sec	8, 9, 3	1.08



Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on `sq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in `[[8, 9]]`.

Runtime	Qubits	Max prob
4.89 sec	8, 9	0.839



Runtime	Qubits	Max prob
2.36 sec	2, 3	0.89



10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on `siq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in `[[8, 3], [8, 9]]`.

Runtime	Qubits	Max prob
2.95 sec	9, 3, 8	0.265



Runtime	Qubits	Max prob
2.43 sec	0, 2, 3	0.385



11. GHZ state preparation

GHZ circuit with 3 qubits executed on `siq20` backend with 1000 shots.

Runtime	Qubits	Success
3.33 sec	3, 8, 9	0.497



Runtime	Qubits	Success
3.95 sec	0, 3, 2	0.910

